

Engineering Nonclassical States via the Dynamical Casimir Effect

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Nonadiabatic driving in ultrastrongly coupled light-matter systems is commonly regarded as a source of errors, as counter-rotating interactions convert vacuum fluctuations into real excitations through the dynamical Casimir effect (DCE). Here we show that, instead, the DCE can be harnessed as a resource for engineering nonclassical states of light. Considering a cavity mode ultrastrongly coupled to a frequency-tunable qubit, we employ optimal quantum control to design driving protocols that convert vacuum fluctuations into targeted states. Numerical optimization reveals a versatile and robust approach for the deterministic preparation of a broad class of nonclassical states, illustrated here through Fock states, squeezed states, and Schrödinger-cat-state superpositions.

Introduction.—The generation of nonclassical states of light is a key objective in quantum technologies, with applications in quantum information processing and quantum-enhanced sensing [1, 2]. Squeezed states enable measurements beyond the standard quantum limit [3, 4], as exemplified by gravitational-wave detectors [5], whereas Fock states constitute a fundamental resource for bosonic quantum computing [6]. Recent studies have shown that bosonic modes initialized in Fock states can even provide the energy required to implement arbitrary quantum gates through energy-recycling mechanisms [7]. More generally, superpositions of Fock states and Schrödinger-cat states form the basis of several bosonic error-correction and logical-encoding schemes [8–11].

The ability to rapidly prepare nonclassical states before decoherence degrades their quantum properties is a key requirement. Circuit quantum electrodynamics (QED) provides a promising platform for this purpose, allowing access to the ultrastrong light-matter coupling regime (USC) [12–14], where the interaction strength becomes comparable to the natural frequencies of the uncoupled subsystems. The resulting fast light-matter dynamics open new opportunities for the efficient engineering of nonclassical states.

However, pushing state-generation protocols into the USC regime inevitably brings the dynamical Casimir effect (DCE) into play [15–20], since nonadiabatic modulation of system parameters generates pairs of excitations from the vacuum. To date, the DCE has been predominantly regarded as an unwanted effect that limits the fidelity of quantum operations [21]. A notable exception is the pioneering proposal to exploit it as a resource for generating entangled states of superconducting qubits [22].

Here, we exploit the DCE as a resource for the fast

and deterministic generation of nonclassical cavity states. To this end, we consider a single-mode electromagnetic cavity coupled to a two-level system (qubit) in the USC regime. In this regime, counter-rotating terms play a crucial role, enabling photon generation from the vacuum through nonadiabatic modulation of the qubit frequency, a manifestation of the parametric DCE [16]. By employing optimal-control techniques [23–26], we design driving protocols that steer the system toward desired nonclassical states by maximizing the final-state fidelity. We illustrate the versatility of the approach through the generation of representative target states, including Fock, squeezed, and Schrödinger-cat-state superpositions. We systematically assess its performance in terms of fidelity and control cost over a broad range of state parameters. Furthermore, we demonstrate its robustness against both classical control noise and quantum dissipation induced by a thermal environment. Our results establish a viable framework for harnessing the DCE as a resource for the fast and robust generation of high-fidelity nonclassical cavity states.

Model.—We consider a high-quality single-mode electromagnetic cavity coupled to a two-level system (qubit) [12, 27], driven by a time-dependent control field acting on the qubit frequency (see Fig. 1). The system is described by the Hamiltonian $H_S(t) = H_R + H_D(t)$, where the cavity-qubit interaction is described by the Rabi Hamiltonian [28–30] (hereafter we set $\hbar = 1$)

$$H_R = \frac{\omega_q}{2}\sigma_z + \omega_c a^\dagger a + g(a^\dagger + a)(\sigma_+ + \sigma_-), \quad (1)$$

and the control Hamiltonian by

$$H_D(t) = \frac{\Omega_D(t)}{2}\sigma_z \quad (2)$$

Here, a ($a^\dagger$) is the bosonic annihilation (creation) oper-

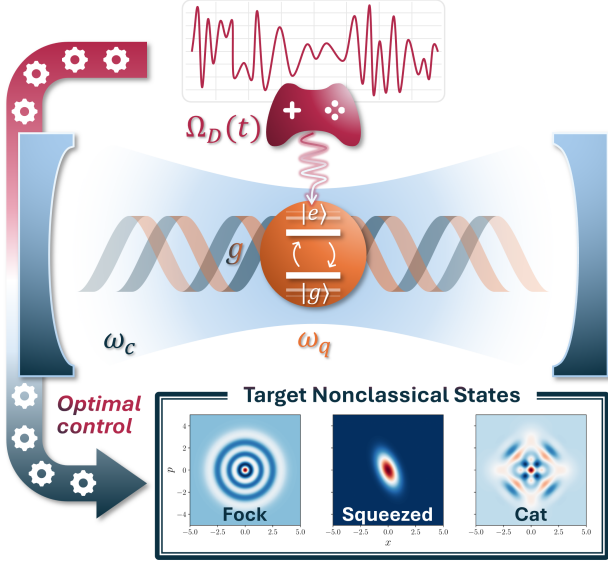


FIG. 1. Schematic representation of a single-mode cavity of frequency ω_c coupled to a qubit of frequency ω_q . An external drive $\Omega_D(t)$ modulates the qubit frequency, inducing dynamical Casimir photon generation in the USC regime. By optimally shaping the drive, target nonclassical cavity states can be generated deterministically from the quantum vacuum.

ator, $\sigma_+ = |e\rangle\langle g|$ ($\sigma_- = \sigma_+^\dagger$) is the qubit raising (lowering) operator, $\sigma_z = |e\rangle\langle e| - |g\rangle\langle g|$ is the Pauli- z operator, ω_c (ω_q) is the cavity (qubit) frequency, g denotes the light-matter coupling strength, and $\Omega_D(t)$ is the external driving classical field. The state $|g\rangle$ ($|e\rangle$) is the ground (excited) state of the qubit. We focus on the USC regime ($g/\omega_c \gtrsim 0.1$) [12–14], where the counter-rotating terms in the cavity-qubit interaction, $a\sigma_- + a^\dagger\sigma_+$, are no longer negligible and contribute significantly to the system dynamics. These terms break the conservation of the total number of excitations, $n_{\text{ex}} = a^\dagger a + \sigma_+ \sigma_-$, allowing photons to be generated from the vacuum via the DCE [15–19, 31] [32]. While excitation number is not conserved, the parity $\Pi = e^{i\pi n_{\text{ex}}}$ is conserved by both the Rabi [30, 33] and the control Hamiltonians thus excitations are created/annihilated in pairs.

Optimal control strategy.— By modulating the instantaneous qubit frequency $\omega_q + \Omega_D(t)$, we harness the DCE to induce cavity excitations in a controlled way. Specifically, we initialize the system in the state $|\psi_0\rangle = |0\rangle|g\rangle$, where $|0\rangle$ denotes the cavity vacuum. In the absence of the counter-rotating terms, this state remains invariant under arbitrary modulations of the qubit frequency. By contrast, in the presence of counter-rotating terms, the modulation can drive the system away from the vacuum and steer it toward nonclassical target states of the cavity, such as Fock states (detailed below), as well as squeezed and Schrödinger-cat-state superpositions (see End Matter). To this end, we employ the framework of quantum

optimal control [23] to shape the time-dependent modulation $\Omega_D(t)$ of the drive (2). Assuming the system evolves from time $t = 0$ to a final time $t = T$, the control objective is to maximize the fidelity F between the final evolved state and the chosen cavity target state. This is achieved by optimizing the control pulse to minimize the cost function defined by the infidelity, $\mathcal{C} = 1 - F$ [34].

We adopt a hybrid optimal control strategy that combines gradient-free and gradient-based methods. First, we use the Chopped Random Basis (CRAB) optimal control technique [24, 25], a gradient-free method. Here, the control field is parametrized as a truncated Fourier expansion,

$$\Omega_D(t) = s(t) \sum_{k=1}^{N_f} [A_k \sin(\omega_k t) + B_k \cos(\omega_k t)], \quad (3)$$

where $\{A_k, B_k\}$ are variational parameters to be optimized, N_f is the number of harmonics, and ω_k are randomly sampled frequencies. This parametrization naturally yields smooth control fields, while the envelope function $s(t)$ ensures smooth turn-on and turn-off of the external drive, enforcing $\Omega_D(0) = \Omega_D(T) = 0$. Additionally, fixing the norm of the Fourier coefficients constrains the total control power, thereby restricting the CRAB optimization to physically realistic regimes. The solution provided by CRAB is then fed as a high-quality initial guess for Gradient Ascent Pulse Engineering (GRAPE) [26], a gradient-based optimal control method. This allows GRAPE to refine the pulse profile to enhance the final fidelity. In GRAPE, the total evolution time $[0, T]$ is discretized into N_t intervals of duration $\delta t = T/N_t$, and the control field is represented as a piecewise-constant function, $\Omega_D(t) \rightarrow \{\Omega_D^{(1)}, \Omega_D^{(2)}, \dots, \Omega_D^{(N_t)}\}$. The control amplitudes $\{\Omega_D^{(k)}\}$ are iteratively updated using analytical gradients of the cost function, enforcing amplitude constraints to ensure compliance with the imposed power limitations. This GRAPE-based refinement step substantially enhances the fidelity. A final interpolation step is applied to smooth the control field, yielding a continuous and more experimentally feasible pulse with a negligible loss in fidelity. Additional technical details on the optimization procedure are provided in the Supplemental material (SM).

Finally, inspired by counterdiabatic-driving methods [35, 36], to quantify the energetic cost of the control fields for the nonclassical-state generation we compute the following state- and setup-independent cost functional

$$C_2(t) := \int_0^t ds \|H_D(s)\|_2^2 \quad (4)$$

for the drive Hamiltonian (2), where $\|X\|_2 = \sqrt{\text{Tr}[X^\dagger X]}$ denotes the Frobenius norm [37].

Fock-state generation.— To demonstrate the effectiveness of our protocol, we tackle the generation of a highly

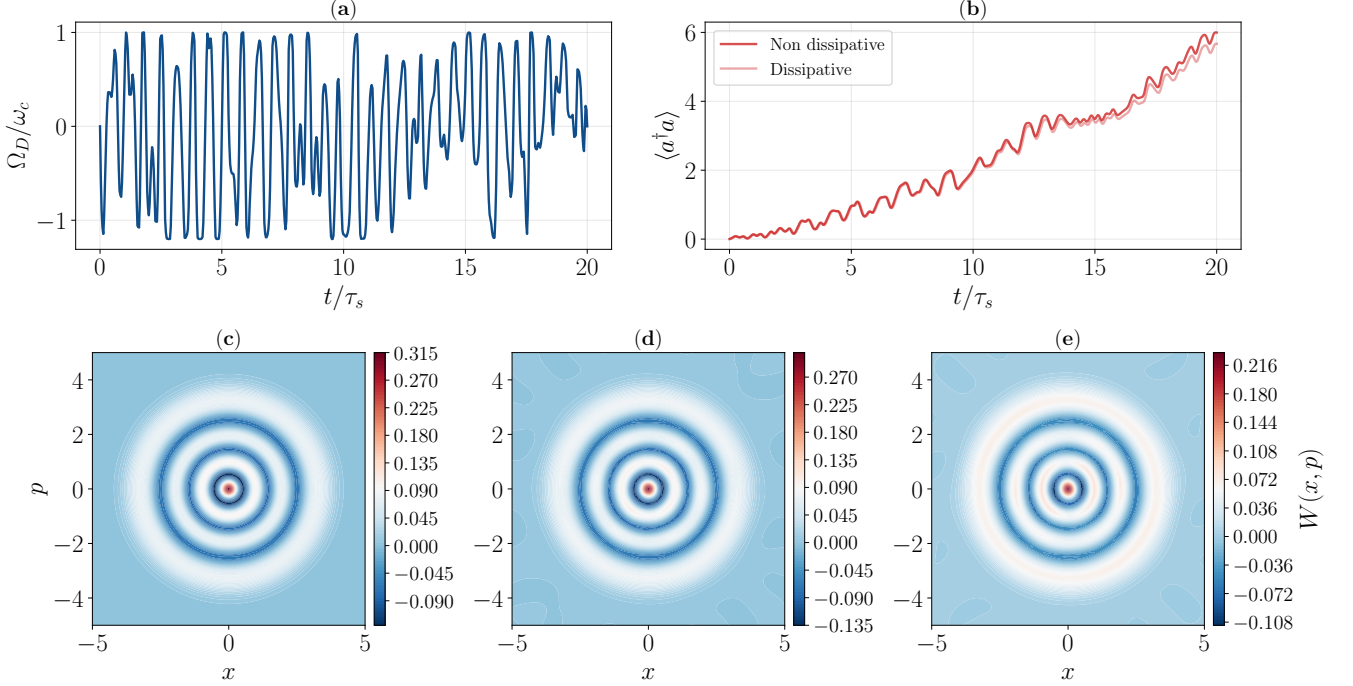


FIG. 2. Optimal control strategy for the generation of the Fock state $|n = 6\rangle$. (a) Optimized control drive $\Omega_D(t)/\omega_c$ obtained via CRAB optimization, GRAPE refinement, and final interpolation, and (b) corresponding cavity mean photon number $\langle a^\dagger a \rangle(t)$. (c-e) Wigner functions of the cavity field comparing (c) the ideal target state $|n = 6\rangle$ with the state prepared by the optimized drive in (a) under (d) unitary dynamics (fidelity $F = 0.995$) and (e) open-system dynamics (fidelity $F = 0.901$). The phase-space distributions are plotted as a function of the field quadrature operators, defined as $x = (a + a^\dagger)/\sqrt{2}$ and $p = (a - a^\dagger)/i\sqrt{2}$. System and control parameters are $g/\omega_c = 0.3$, $T = 20\tau_s \equiv 20\pi/2g$, $\omega_q/\omega_c = 2$ (corresponding to the off-resonance regime $|\omega_c - \omega_q| \gg g$), and $N_f = 20$. The number of time intervals for the GRAPE refinement is $N_t = 300$. Environmental parameters are $\omega_B/\omega_c = 10$, $\beta\omega_c = 5$, and $\eta = 10^{-4}$ (see section *Robustness against noise* for details).

nonclassical Fock state with $n = 6$ as a first benchmark (see Fig. 2). Starting from the vacuum, the optimized control drive [panel (a)] steers the cavity population [panel (b)] to successfully prepare the desired $|n = 6\rangle$ state with a fidelity of $F \approx 0.995$ over a total evolution time $T = 20\tau_s$, where $\tau_s = \pi/2g$ sets the characteristic timescale [38]. The excellent accuracy of the protocol is further confirmed by the final state's Wigner function, which faithfully reproduces the alternating, concentric interference rings characteristic of the ideal target state [panels (c) and (d)] [39].

To evaluate how the protocol performance scales with increasing photon number, we investigate the generation of Fock states over the range $1 \leq n \leq 10$. We first consider bare CRAB optimization, without GRAPE refinement or constraints on the control amplitudes. The protocol performance is assessed in terms of the final-state fidelity and the control cost defined in Eq. (4). For each n , both quantities are averaged over 10 independent CRAB optimizations with different random seeds (see Fig. 3). The average fidelity is consistently high, $F \gtrsim 0.96$, across all target states [panel (a)], confirming the robustness of the control strategy beyond the low-

excitation regime. As n increases, however, the average fidelity gradually decreases, while its dispersion widens, indicating an increasingly intricate control landscape for higher photon numbers. Consistently, the cost [panel (b)] mildly increases with n , reflecting the greater energetic resources required to target highly excited Fock states.

Figure 3 also presents the results obtained by taking, for each n , the highest-fidelity CRAB solution as an initial guess for a constrained GRAPE refinement, followed by pulse smoothing. Notably, the refined fidelity systematically outperforms the bare CRAB average, particularly in the large- n regime. Crucially, the control cost is drastically reduced, proving that amplitude bounds effectively guide the protocol toward far more energetically efficient solutions.

Robustness against noise.— We assess the robustness of the optimized control strategy from Fig. 2 against both classical noise affecting the control field and quantum noise induced by coupling to an external environment.

As *classical control noise*, we consider two representative additive noises $[\Omega_D(t) \rightarrow \Omega_D(t) + \delta\Omega_D(t)]$: (i) white noise $\delta\Omega_D(t) = \gamma\xi(t)$, described as an uncorre-

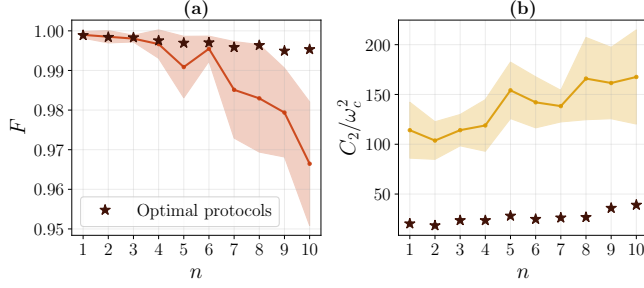


FIG. 3. Performance scaling of Fock-state generation with the target photon number n . Average (a) fidelity, and (b) control cost (4) as a function of n . Markers (mean value) and error bands (standard deviation) are obtained from 10 independent unconstrained CRAB optimizations with different random seeds; the star markers indicate the performance of the optimal protocols obtained by refining the best CRAB solutions through a subsequent GRAPE optimization followed by a final pulse-smoothing interpolation step. The number of time intervals for the GRAPE refinement is $N_t = 400$, the other parameters are the same as those used in Fig. 2.

lated Gaussian process $\xi(t)$ with zero mean and unit variance, and (ii) colored noise $\delta\Omega_D(t) = \gamma\tilde{\xi}(t)$, obtained by filtering white noise through a Gaussian kernel G_{τ_c} with correlation time τ_c , yielding a correlated process $\tilde{\xi}(t) = (\xi * G_{\tau_c})(t)$. The noise amplitude is parametrized by γ . These additive noises capture broadband fluctuations and finite-bandwidth noise, respectively. For each γ , we average the fidelity over 100 independent stochastic realizations of the noise, and show it against the noise amplitude in Fig. 4(a). The protocol remains highly robust against both noise models, maintaining fidelities $F \gtrsim 0.95$ up to $\gamma \sim 0.1$ for white noise, and up to $\gamma \sim 0.2$ for colored noise. The relatively higher robustness against colored noise can be understood from the broadband nature of white noise, whose high-frequency components effectively disrupt the dynamics.

Turning to *quantum noise*, we couple the the cavity-qubit system to a thermal bosonic environment at inverse temperature β . The environment is modeled as a bath of harmonic oscillators with an Ohmic spectral density with exponential cutoff, $J(\omega) = \eta|\omega|e^{-|\omega|/\omega_B}$, where η sets the system-bath coupling strength and ω_B is the bath cutoff frequency. The reduced system dynamics are described within the Gorini-Kossakowski-Lindblad-Sudarshan (GKLS) formalism [40–42], valid under the assumptions of weak system-bath coupling and Markovian dynamics. The evolution of the system density matrix $\rho_S(t)$ is governed by the GKLS master equation

$$\frac{d}{dt}\rho_S(t) = -i[H_S(t), \rho_S(t)] + \sum_{\omega} \left(L_{\omega}(t)\rho_S(t)L_{\omega}^{\dagger}(t) - \frac{1}{2}\{L_{\omega}^{\dagger}(t)L_{\omega}(t), \rho_S(t)\} \right), \quad (5)$$

where the Lindblad operators $L_{\omega}(t)$ account for the dis-

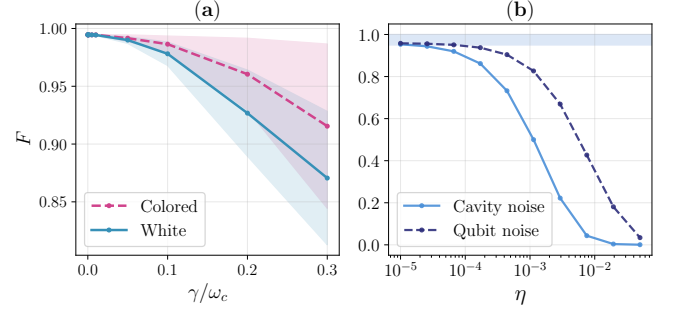


FIG. 4. Robustness of Fock-state generation against classical and quantum noise. (a) Classical control noise: Fidelity as a function of the dimensionless noise amplitude γ/ω_c for two additive noise models (white noise, and colored noise with finite correlation time $\tau_c = \tau_s/3$). Markers (mean value) and error bands (standard deviation) are obtained from 100 independent realizations. (b) Quantum noise: Fidelity as a function of the system-bath coupling strength η for Lindblad operators acting on the cavity (solid curve) and on the qubit (dashed curve). The high-fidelity region ($0.95 \leq F \leq 1$) is highlighted for reference. System parameters are the same as those used for the optimized control pulse in Fig. 2. Environmental parameters (see SM) are $\omega_B/\omega_c = 10$ and $\beta\omega_c = 5$.

sipative channels, acting either on the cavity or on the qubit (see SM for details on the microscopic model). Figure 4(b) characterizes the robustness of the optimal $|n=6\rangle$ Fock-state generation from Fig. 2 as a function of the system-bath coupling strength η . The fidelity remains remarkably high ($F \gtrsim 0.9$) up to $\eta \lesssim 10^{-4}$ (with qubit noise exhibiting consistently higher fidelity values compared to cavity noise), before rapidly decreasing at larger noise amplitudes. At this representative threshold $\eta = 10^{-4}$, the state remains highly faithful to the target Fock state: as shown by the Wigner function in Fig. 2(e) [see also panel (b), where the mean photon number obtained in the dissipative case closely tracks the corresponding nondissipative evolution], the protocol yields a fidelity of $F = 0.901$, compared with the ideal value of $F = 0.995$ obtained under unitary, noise-free evolution.

Conclusions and discussion.—We have shown that the DCE in a cavity-qubit system operating in the USC regime can be harnessed for the high-fidelity generation of nonclassical cavity states. By employing optimal-control techniques, we design driving protocols that modulate the qubit frequency and convert vacuum fluctuations into controllable excitations, thereby enabling the preparation of target states. Furthermore, our hybrid optimization strategy naturally accommodates constraints on the control amplitudes, limiting the overall control power and facilitating adaptation to experimental requirements.

The generation of Fock states $|n\rangle$ is remarkably effective, yielding fidelities exceeding 0.99 up to $n = 10$, while remaining highly robust against both classical con-

trol noise and quantum dissipation. Importantly, the approach extends beyond Fock-state preparation to a broad class of nonclassical states, as illustrated by squeezed-vacuum states and Schrödinger-cat-state superpositions (see End Matter).

The proposed protocol represents a promising route toward the realization of ultrafast, robust, high-fidelity nonclassical-state generation in circuit-QED platforms operating in the USC regime [43, 44]. The optimized control pulses can be implemented using currently available microwave-control technology [45], while the open-system simulations here discussed employ system/bath parameters representative of contemporary superconducting devices [46, 47]. Moreover, the USC regime enables state-preparation times substantially shorter than those achieved in weak-coupling architectures. Additional details and quantitative benchmarks are provided in the SM.

Overall, our findings establish the DCE as a resource—rather than a limitation—for quantum-state engineering. The impact of the proposed protocol is expected to be particularly significant in applications where state preparation must be deterministic rather than heralded [48]. These include quantum computing, ranging from quantum gate operations [7] and continuous-variable encodings [49] to quantum error correction [8, 11, 50, 51], as well as quantum communication [52, 53] and quantum metrology [54], where nonclassical states must be generated on demand and, in many cases, at high repetition rates.

More broadly, extending the optimal-control framework developed here to multi-qubit cavity architectures may open new avenues for scalable quantum-state engineering. The large Hilbert spaces and intrinsic nonlinear dynamics of such systems also make them attractive candidates for quantum reservoir computing [55–58].

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END MATTER

Appendix A: Squeezed state generation.—To illustrate how our optimal control framework generalizes to other nonclassical states, we focus on the preparation of single-mode squeezed vacuum states of the cavity field, defined as $S(r, \theta) |0\rangle$, where

$$S(r, \theta) = \exp \left[\frac{1}{2} r (e^{-i\theta} a^2 - e^{i\theta} a^{\dagger 2}) \right] \quad (\text{A1})$$

is the squeezing operator, $r \geq 0$ the squeezing strength, and θ the squeezing angle. Squeezed states are characterized by reduced fluctuations in one field quadrature at the expense of increased fluctuations in the conjugate one. For a single-mode squeezed vacuum, the variance of the squeezed quadrature is $\Delta X^2 = e^{-2r}/2$, where $\Delta X_0^2 = 1/2$ is the vacuum variance. In experiments, squeezing is typically quantified in decibels (dB) as

$$r_{\text{dB}} = -10 \log_{10} \left(\frac{\Delta X^2}{\Delta X_0^2} \right) = 20 r \log_{10}(e), \quad (\text{A2})$$

which can be inverted as $r \simeq r_{\text{dB}}/8.686$.

To determine the optimal control protocol, we employ the same control parametrization, $\Omega_D(t)$, and multi-stage optimization pipeline (CRAB, GRAPE, and final interpolation) developed for Fock-state generation, where the infidelity cost functional is now defined with respect to the target squeezed state. As a first benchmark, we focus on the generation of a mildly squeezed state ($r_{\text{dB}} = 3$) with a finite squeezing angle ($\theta = \pi/4$), resulting in a rotated squeezing axis in phase space. The drive obtained from this optimization [Fig. 5(a)] prepares a state that achieves an exceptional fidelity of $F \simeq 0.999$ with respect to the target single-mode squeezed vacuum state, as reflected in the excellent agreement between the Wigner functions [Figs. 5(c,d)].

Then, we investigate the protocol's performance across experimentally attainable squeezing strengths $r_{\text{dB}} \in [0, 16]$ [60–63], setting $\theta = 0$ for simplicity. For each squeezing strength r , we run 10 independent bare CRAB optimizations (imposing no constraints on the control amplitudes) with different random seeds to evaluate the average fidelity and control cost [Fig. 6(a,b)]. For weak to moderate squeezing, the protocol consistently yields high fidelities ($F \gtrsim 0.98$), performing comparably to Fock-state generation. At stronger squeezing, the average fidelity gradually degrades as r increases, accompanied by larger fluctuations across different optimization trials that suggest an increasing complexity of the control landscape. Concurrently, the control cost is nearly constant in the weak squeezing regime, while increasing mildly at stronger squeezing. However, because this trend remains comparable to the statistical fluctuations across different realizations, no definitive scaling can be resolved within the error bars.

Fig. 6(a,b) also shows the results obtained by selecting the best CRAB solutions (in terms of fidelity) for each r and refining them via GRAPE under explicit amplitude bounds, followed by pulse smoothing. The refined fidelity slightly outperforms the bare CRAB average for all r , while the corresponding control cost is drastically reduced.

Overall, while no definitive scaling of the required control resources emerges, the results suggest that strongly squeezed states demand more structured control fields and exhibit larger variability in the optimization outcomes. Nevertheless, the protocol remains effective across the explored range of squeezing, demonstrating its versatility for nonclassical state generation.

Appendix B: Generation of Schrödinger-cat-state superpositions.—We further assess the versatility of our protocol by targeting superpositions of Schrödinger-cat states. The standard two-component even (+) and odd (−) Schrödinger-cat states are defined as

$$|C_{\alpha}^{\pm}\rangle \propto |\alpha\rangle \pm |-\alpha\rangle, \quad (\text{B3})$$

where $|\alpha\rangle$ denotes a coherent state, satisfying $a|\alpha\rangle = \alpha|\alpha\rangle$, with $\alpha \in \mathbb{C}$. The complex parameter α represents the displacement in the phase space, and its squared magnitude $|\alpha|^2$ corresponds to the average number of photons in each coherent component.

As an illustrative and more challenging benchmark, we test the protocol for the generation of a superposition of cat states, constructed along two orthogonal axes of the phase space:

$$|\psi_{\text{target}}\rangle = \mathcal{N} (|C_{\alpha}^{+}\rangle - |C_{i\alpha}^{+}\rangle), \quad (\text{B4})$$

where $\mathcal{N}$ is the appropriate normalization constant, and restricting to $\alpha \in \mathbb{R}$. This specific state features a highly non-classical checkered pattern of positive and negative interference fringes at the center of the Wigner function, making it extremely sensitive to phase-space displacements. We employ the same control parametrization, $\Omega_D(t)$, and multi-stage optimization pipeline (CRAB, GRAPE, and final interpolation) developed for Fock and squeezed-state generation, where the infidelity cost functional is now defined with respect to the target cat state.

Specifically, targeting the superposition state with $\alpha = 2$ (where each coherent component carries an average of $|\alpha|^2 = 4$ photons), the protocol achieves a remarkably high final fidelity of $F = 0.978$; the corresponding optimized drive and Wigner functions are presented in Fig. 5(b,e,f), showcasing an almost perfect agreement between the target [Fig. 5(e)] and reconstructed [Fig. 5(f)] states. We emphasize that the protocol achieves a significantly reduced time-to-solution compared to standard gate-based approaches (see SM for details).

Finally, we investigate how the protocol's performance in generating the target state in Eq. (B4) scales with

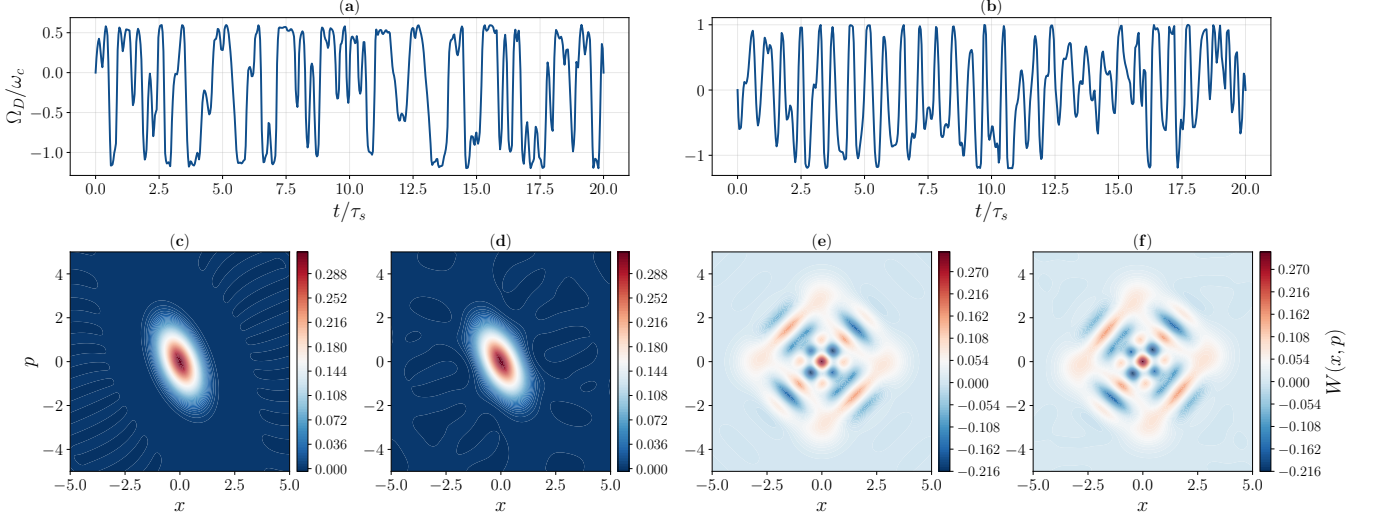


FIG. 5. Optimal control strategy for the generation of squeezed and cat states. (a, b) Optimized control drives $\Omega_D(t)/\omega_c$ used for the generation of (a) the squeezed vacuum state and (b) the superposition of Schrödinger-cat states in Eq. (B4). (c-f) Wigner functions of the cavity field comparing the ideal target states (c, e) with the final states obtained via optimal control (d, f). Specifically, (c, d) refer to the squeezed vacuum state with squeezing strength $r = 3$ dB and squeezing angle $\theta = \pi/4$, while (e, f) refer to the Schrödinger-cat state with $\alpha = 2$. The number of time intervals for the GRAPE refinement is $N_t = 400$, the other parameters are the same as those used in Fig. 2.

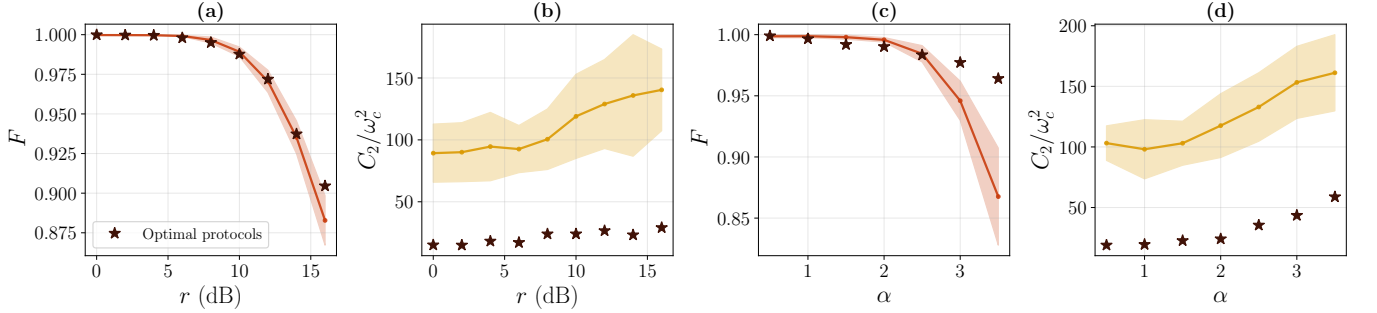


FIG. 6. Performance scaling of squeezed and cat-state generation. (a, b) Squeezed-vacuum state generation as a function of the squeezing strength r_{dB} at $\theta = 0$. (c, d) Generation of Schrödinger-cat-state superpositions as a function of α [cf. Eq. (B4)]. We show the average (a, c) fidelity, and (b, d) control cost C_2 (4) associated with the driving. Markers (mean value) and error bands (standard deviation) are obtained from 10 independent unconstrained CRAB optimizations with different random seeds. The star markers indicate the performance of the optimal protocols obtained by refining the best CRAB solutions through a subsequent GRAPE optimization followed by a final pulse-smoothing interpolation step. The number of time intervals for the GRAPE refinement is $N_t = 600$ (for squeezed states) and $N_t = 400$ (for cat states), the other parameters are the same as those used in Fig. 2.

α . For each value of α , we run 10 independent unconstrained CRAB optimizations with different random seeds, and evaluate the fidelity and control cost, reporting their mean value and standard deviation in Fig. 6(c,d). The fidelity is consistently high ($F \gtrsim 0.95$) for $\alpha < 3$, and degrades for larger α [Fig. 6(c)]. The control cost shows an overall increasing trend as α increases [Fig. 6(d)].

Fig. 6(c,d) also shows the results obtained by selecting the best CRAB solutions (in terms of fidelity) for each α and refining them via GRAPE under explicit amplitude bounds, followed by pulse smoothing. Notably, while this refined fidelity remains very high at low α , it significantly outperforms the bare CRAB average for $\alpha > 2$. Consistent with our observations for Fock and squeezed states, the corresponding control cost is drastically reduced.